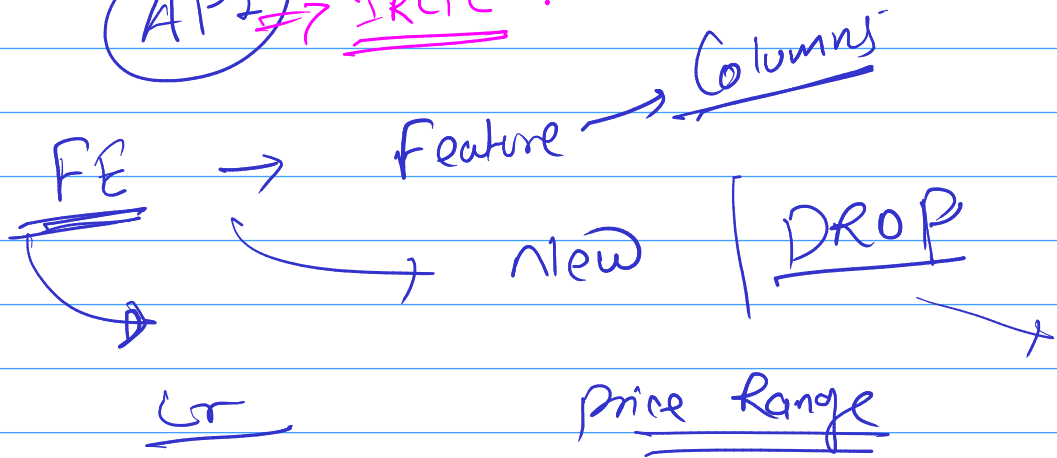


Cloud → www.dramwithdata.in

API → IRCTC .



0 to 1000
1000 to 5000

Statistics

Statistics is a branch of Mathematics that involves Collecting, Analysing, Interpreting and presenting the data.

Why?

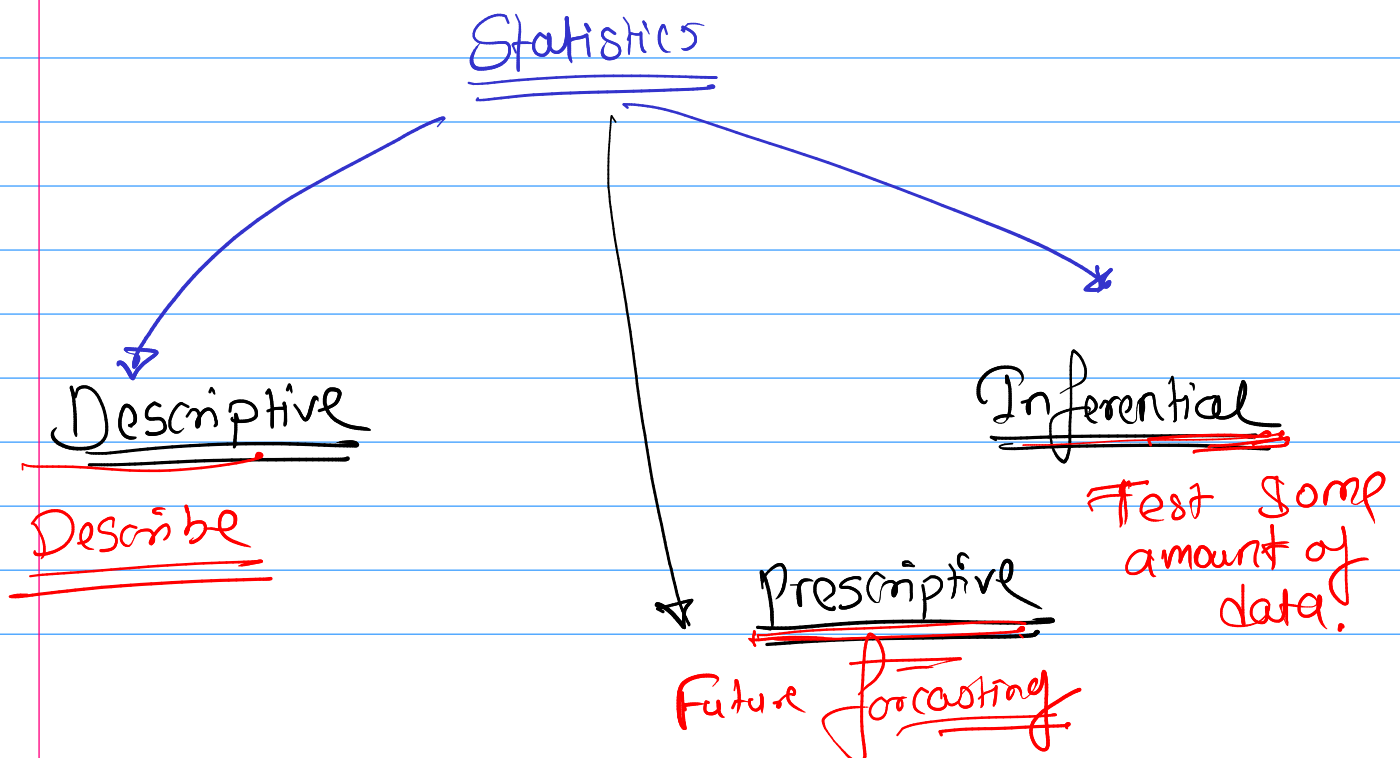
large amount

⇒ Covid-19

↓
6GB

It provide us tools and methods to understand the large amount of data.

Everywhere ⇒ Business, Gov. policies, Finance, Medical, Weather,



Descriptive $\Rightarrow$ It deals with the Collection,
Organization, analysis, interpretation &
presentation of data.

ex data

Inferential $\Rightarrow$ It deals with making Conclusion
and predictions about a population based
on Sample.

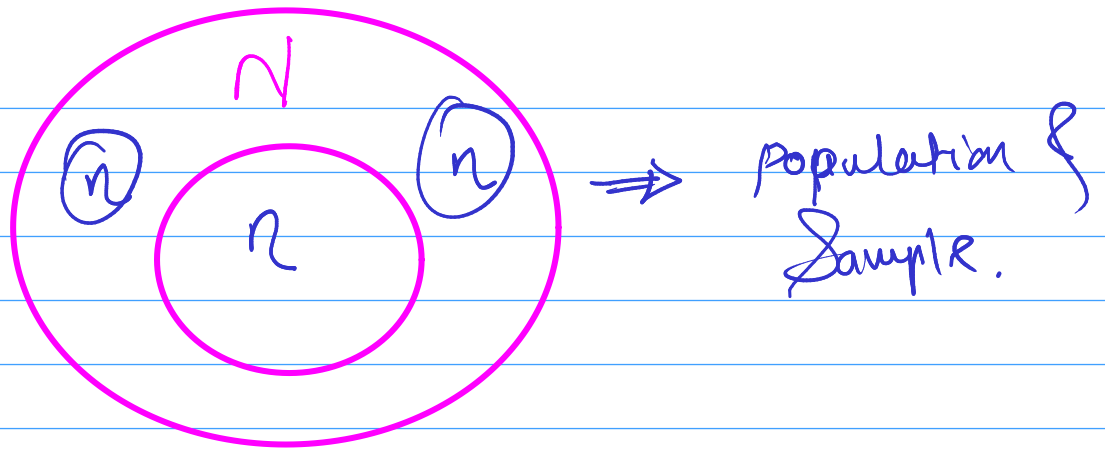
ex $\Rightarrow$ size of snake / Anatonda

Prescriptive $\Rightarrow$ Its Go beyond descriptive
and Inferential to recommend optimal
action.

ex $\Rightarrow$ youtube $\begin{cases} \rightarrow \text{excel} \rightarrow \text{mysql} \\ \rightarrow \text{python} \rightarrow \text{PowerBI} \end{cases}$

Population $\Rightarrow$ It is a complete set of individual,
(N) object, are of interest.
 $\Rightarrow$ Blood in the body.

Sample $\Rightarrow$ It is the subgroup of the
(n) Population.
 $\Rightarrow$ Sample blood for testing.



Type of Data.

Categorical / Qualitative

Numerical / Quantitative

Nominal

No Rank

Male

Female

Gender

Ordinal

Rank

Train Coach

GN	3A
SL	4A
3AC	3A
2AC	2A
1AC	1A

Winner

1st / 2nd / 3rd / runner up

Discrete

Whole No

Age :-

Roll no :-

mo. no. add. no.

no. of children / family member

Continuous

percentage / miles

Height
Weight

Km

(100%) (percentage)

Ratio

⇒

70% 30%

/ Drugs

Interval / Time-Series data

— Weather

→ share Market →

→ gold price

Stop

(No-Zero)

Time /
✓ Hos / min / sec
Date
✓ Date / month / year ✓

Structured & ⇒ Rows & Column

Unstructure ⇒ Text

— x —

Measures of Statistics

①

Center

① Measure of Central Tendency

② Measure of Dispersion → spread.

③ Probability

—————x—————

① Measure of Central Tendency

① Mean (Average)

The mean is Sum of all value in the dataset divided by the number of value.

$$x = [\underset{x_1}{1}, \underset{x_2}{2}, \underset{x_3}{3}, \underset{x_4}{4}, \underset{x_5}{5}] = \text{mean} = \frac{1+2+3+4+5}{5} = \frac{15}{5} = 3$$

$$\text{Sample mean } (\bar{x}) = \frac{\sum_{i=1}^n x_i}{n}$$

$$\text{Populated mean } (\mu) = \frac{\sum_{i=1}^N x_i}{N}$$

① ex $\Rightarrow$ 100 students data in 12th class, where 4 students Age is missing.

[17, 18, 19, 20]

$\Rightarrow$ mean is used to replace the missing value (null value) in the dataset (If there are no "outliers")

∴ ⇒ Because mean is affected by the outliers.

17, 18, 19, 19, 18, 19, 18,

$$[1, 2, 3, 4, 5, 6666] = \frac{1+2+3+4+55}{5} \quad \text{outlier}$$
$$= \frac{65}{5} = \underline{13}$$

② Median ⇒ It is not affected by the outlier.

⇒ It is the middle value in the dataset but after Sorting.

$$55, 4, 1, 2, 3 = \frac{55+4+1+3+2}{5} = 13$$

55, 4, 3, 2, 1

1, 2, 3, 4, 55

If there is an outlier in the numerical data,

we replace missing values with median.

Repetable

Q $\Rightarrow$ It is used for Categorical data to replace the missing value.

~~missing~~

$$m = 36$$

mode = f

- file have two mode

- file have two mode

DS
AI

④ Weighted Mean $\Rightarrow$ (or dekho)

The weighted mean is the sum of all products of each value and its weight, divided by the sum of weight.

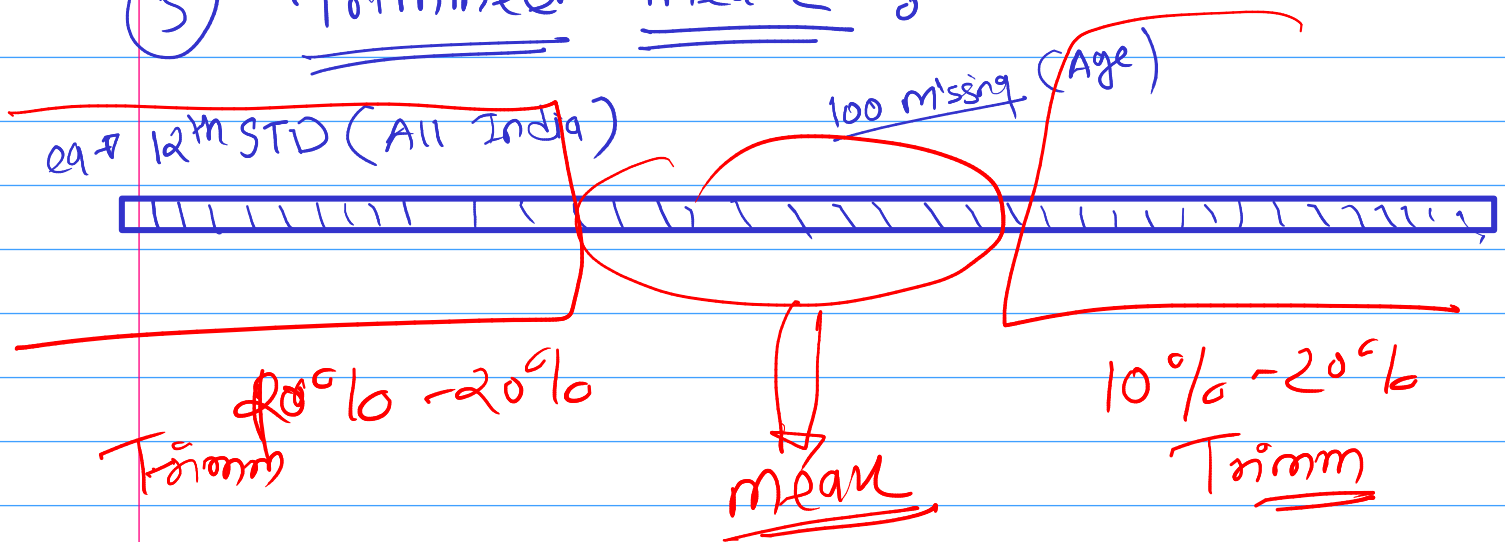
ML

$$\boxed{LR} \underline{0.2} = \underline{\underline{10h}}$$
$$\boxed{AF} \cdot 0.3 = 15h$$

XGBoost 0.4 = 17L

$$\begin{array}{r} 10 \times 0.2 + 15 \times 0.3 \\ + 17 \times 0.4 \\ \hline 0.2 + 0.3 + 0.4 \end{array}$$

5) Trimmed Mean :-



It is calculated by removing a certain percentage of the smallest and largest value from the dataset and then taking a mean of remaining value.